



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

CS6903: Network Security (Intrusion Detection Systems)



Saurabh Kumar
CSE Department
IIT Hyderabad



Acknowledgement

- ❑ Prof. Sandeep Kumar Shukla, IITK (Now at IIIT Hyderabad)
- ❑ Jason Crossland, Johns Hopkins University
- ❑ Various other sources on internet

Introduction

❑ What is Intrusion Detection?

- Monitoring network/system activities for malicious actions
- Key component of defense-in-depth strategy

❑ Why IDS is Important?

- Detect unauthorized access
- Alert on suspicious/malicious activity
- Support forensic analysis (Note: It is not a forensics tool)

Formally Defining

*“In Information Security, intrusion detection is an act of **detecting** actions that attempt to **compromise the confidentiality, integrity or availability** of a resource”*

Wikipedia

Types of IDS

❑ Network-based IDS (NIDS)

- Monitors network traffic
- Good for perimeter and broad visibility
- E.g., Zeek, Snort, Suricata

❑ Host-based IDS (HIDS)

- Monitor OS logs, files, processes
- Detect insider misuse, privilege escalation
- E.g., OSSEC, Tripwire

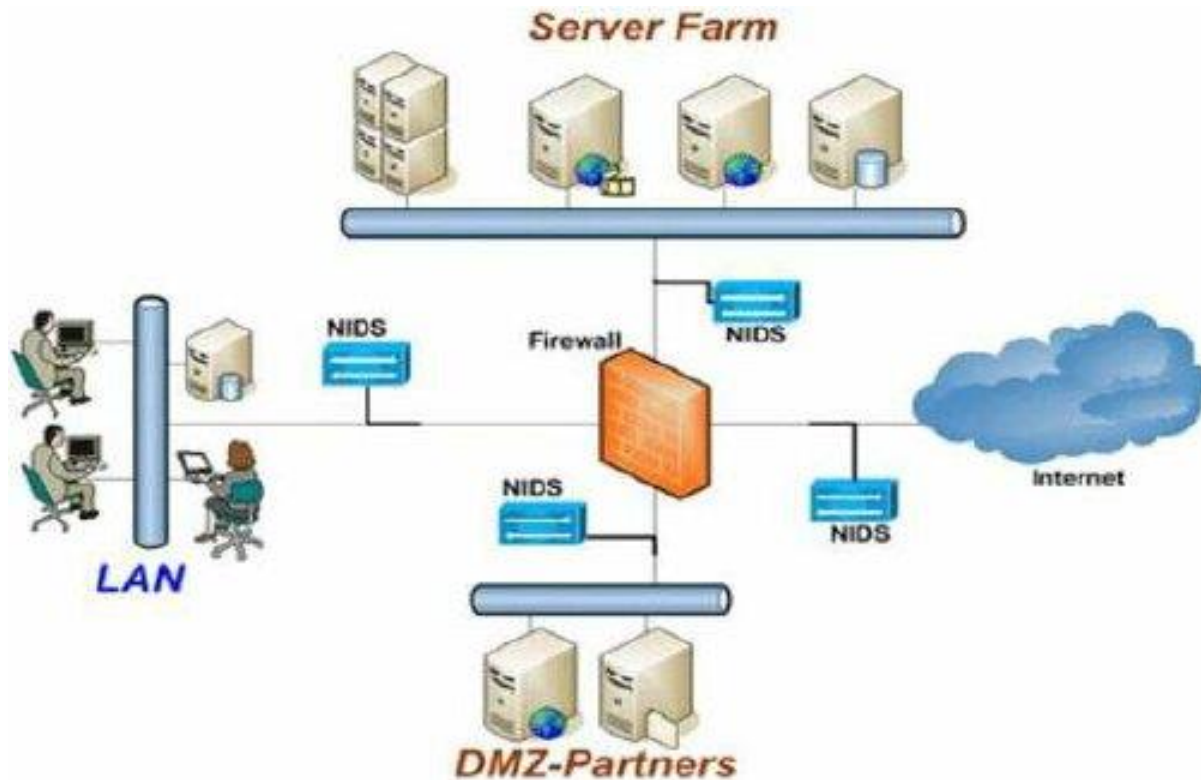
❑ Hybrid IDS

- Combination of NIDS + HIDS

❑ (Application) Protocol specific IDS

- Like IDS for web application

Where to Place NIDS



NIDS: Before Firewall

❑ Where?

- Between the internal network and external (Internet)

❑ Advantage:

- Monitor all incoming and outgoing traffic
- First line of defense, can catch many attacks before they enter
- Centralized monitoring point, easy to manage

❑ Disadvantages:

- Encrypted traffic (HTTPS, VPN) may bypass detection
- Cannot detect lateral (internal-to-internal) attacks
- High traffic volume may cause packet drops.

NIDS: After Firewall (Just after)

❑ Where?

- Just after the firewall, on the internal network side

❑ Advantage:

- Sees what traffic actually passes through firewall rules
- Can detect attacks that bypass misconfigured firewall rules

❑ Disadvantages:

- Limited visibility of dropped/blocked packets
- May miss reconnaissance attempts stopped at firewall

NIDS: After Firewall (At Critical Network Segment)

❑ Where?

- Between sensitive network zones (e.g., user LAN ↔ server farm)

❑ Advantage:

- Can detect lateral movement and insider threats
- Protects critical assets directly

❑ Disadvantages:

- Needs multiple sensors if many critical zones exist
- Higher cost and management complexity

NIDS: After Firewall (On Server Farm / Data Center Ingress/Egress)

❑ Where?

- In front of servers hosting applications (DB, Web, Email)

❑ Advantage:

- Directly monitors high-value targets
- Easier to detect anomalies or DoS attempts against servers

❑ Disadvantages:

- Limited view of other parts of the network
- High-speed links in data centers require powerful NIDS hardware

NIDS: After Firewall (Inside the Internal Network: User LAN / Access Layer)

❑ Where?

- Near switches that connect users

❑ Advantage:

- Detects worm outbreaks, compromised machines, insider misuse
- Helps track lateral movement

❑ Disadvantages:

- Requires many sensors across multiple access points
- High operational complexity

Distributed/Hybrid Deployment

❑ Where?

- Combination of perimeter + internal + server-side placements

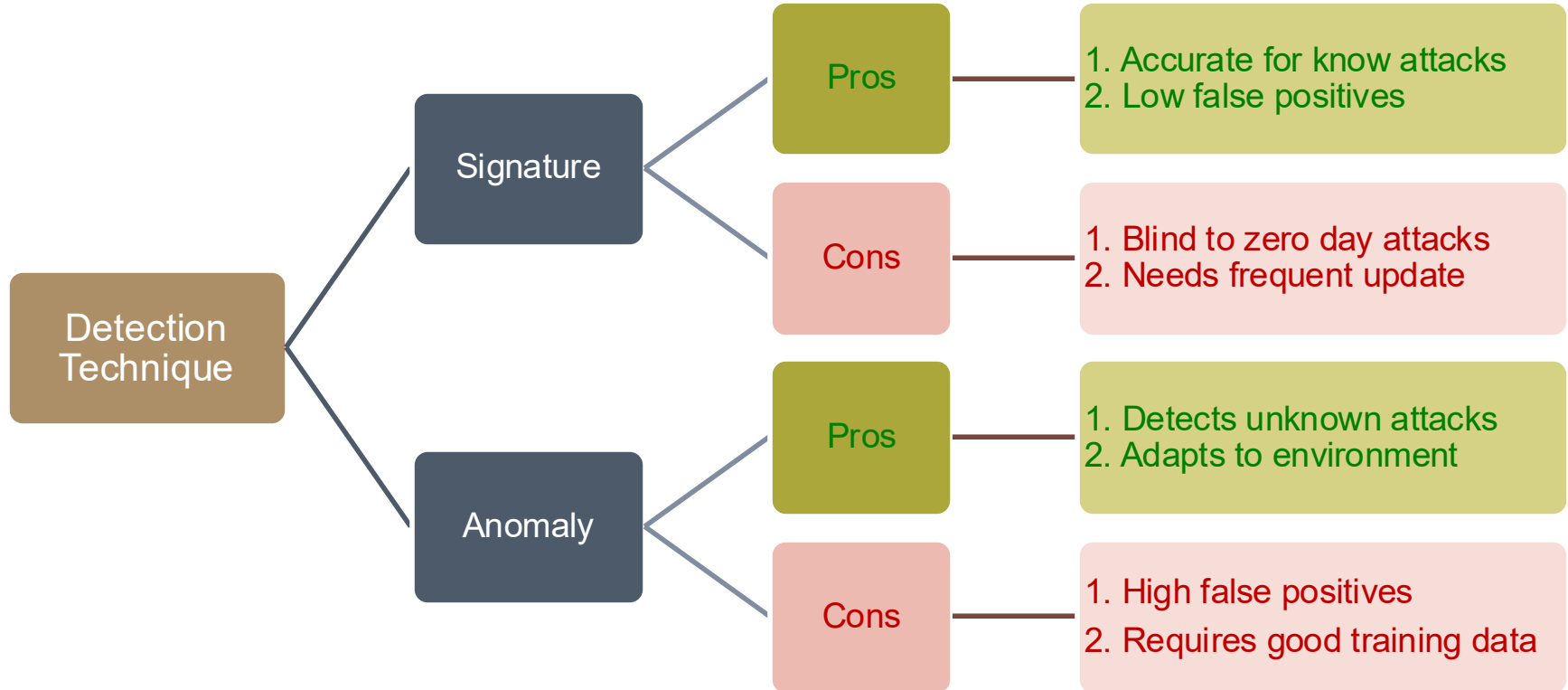
❑ Advantage:

- Comprehensive coverage (external + internal + lateral)
- Flexible and resilient to evasion

❑ Disadvantages:

- Expensive, requires skilled monitoring team.
- Can lead to alert overload if not fine-tuned

Detection Techniques



Best Practices

- ❑ Use both NIDS and HIDS
- ❑ Update signatures and train anomaly models
- ❑ Tune rules to reduce noise
- ❑ Place sensors strategically